FOREWORD

This text attempts to present an application-oriented introduction to information theory, with less theorems and proofs than usual, and more engineering considerations. The approach and notation is very similar to that of the excellent textbook [1], with some simplifications and extensions that seemed necessary for a single-semester university course. For those students who have an interest in the more detailed mathematical background of the material, the books [2] and [3] are recommended for further reading.

[1] Richard B. Wells: *Applied Coding and Information Theory for Engineers*, Prentice Hall, 1999.
[2] Steven Roman: *Introduction to Coding and Information Theory*, Springer, 1997.
[3] R.B. Ash: *Information Theory*, Dover Publications, 1990.

I SOURCE CODING

1. Entropy and Mutual Information

1.1. The Cocept of Information (Everyday, Philosophical, Engineering)

The philosophical definition of information, the basis of epistemology[1] [2].

- Plato: **Phaedo** (5th century B.C.). *"If we truly wish to know something, we must free ourselves from the body and contemplate through only the soul itself"*. Based on this, the soul with its spiritual vision can see the true meaning of things, the inner **form,** idea, eidos (form, essence) but only after it has freed itself from the body. This theory of pure, direct intellectual understanding probably had great influence on Aristotle.
- Aristotle: **On the Soul**. Plato's student, Aristotle, with this work of his has laid the foundations of how ancient Christian, Jewish and Arabic philosophies described the meaning of the soul itself. In his radical epistemological thesis, he stated that "*the soul, in a certain way, is unified with every living thing*". During the process of comprehension, the deeper part of the human intellect, the so-called passive intellect, alters itself to the subject's **inner form**, to its substance. This alteration is the inner development of the soul (which itself is a form, the inner form of the body), this is **information**.
- Eriugena: **The Division of Nature** (Neoplatonic Christian mystic). By the Christian theological-anthropological aspect **form** is the inner form of humans, which Eriugena considers to be nothing else but Christ himself. Thus mankind's task is to **conform** to this holy inner form.
- Ficino: **Five Questions Concerning the Mind** (end of 15th century). Marsilio Ficino was one of the most influential humanist philosophers of the early Italian Renaissance in the 15th century. His most important work is the *Theologia platonica de immortalitate animorum* (Platonic Theology). Following in the footsteps of Aristotle, Ficino discusses how the soul sees God through a "vision of intellect". This mixture of Aristotelian philosophy of soul and Neoplatonic mysticism states that with this vision the soul

[1] Epistemology: a branch of philosophy that investigates the origin, nature, methods, and limits of human knowledge.
[2] The original and Hungarian versions of the quoted works are available at the lecturer of the course

transforms into God: God as a **form** joins itself to the soul ("tamquam forma … animae sese iungit")

- John Locke: **An Essay Concerning Human Understanding** (1960). His take on metaphysics, based on Descartes' opinion, is dualism. Locke rejects the Aristotelian philosophy of soul and the Platonic idealism. Despite stating that things have no inner form, the word *information* appears several times in Locke's work, however, not in the modern meaning of manipulable data but rather as someone's inner, moral education or development.

News, Data, Information, Knowledge. The everyday definition of information.
Data and news are carriers of information and the amount of conveyed information is said to be "the amount of knowledge we gained by acquiring it". It is also commonly said that information is "processed data". As an intuitive approach to information, we should take a look at the methods of those who trade in information (e.g. newspapers). We can see that the value of news is based on two related elements:

- The surprise factor (how amazing/shocking it is)
- The relevance factor (how much the news affects the recipient)

An example for a news article concerning everyone would be *"Tomorrow the sun will not rise"*. A less relevant information would be that *"John Doe has finally passed his exams."* (of course for his relatives this is a great achievement worthy of being on the cover page of the family paper, would such exist).
Unexpected components increase the value of information: a statement such as *"The student's mess hall will be closed tomorrow due to pest control"* might go as far as the university news, but one like *"The student's mess hall will be closed tomorrow due to a rattlesnake infestation"* could appear in the nationwide media.

From these two factors of information's everyday meaning, the engineering definition and *information theory* apply only the first, the "surprise factor". The fundamentals of this field of study were laid by C.E. Shannon (1916-2001) in 1948.

1.2. Characterising Discrete Sources, Entropy

Basically our aim is to capture and transmit the information through space (e.g. broadcasting) and/or time (e.g. a book, DVD) from a source to a destination (often called sink). The carriers of the information can be diverse, e.g. radio waves, wires, smoke signals, etc. These carriers are usually referred to as the "**channel**". Unfortunately, in practical applications we always have to take the presence of **noise** into account in this channel.

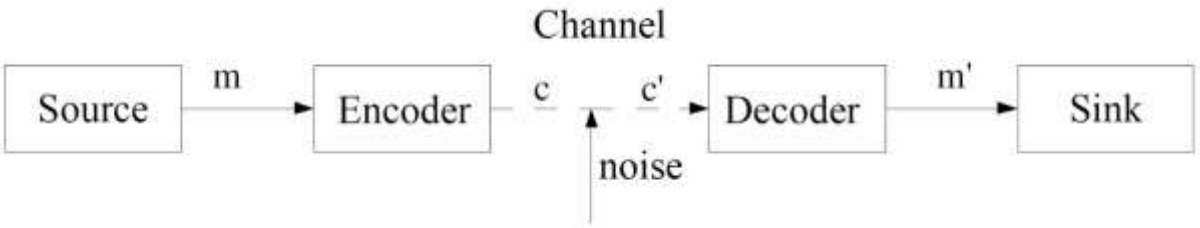


Figure 1.1: *Basic model of information transmission*

In Figure 1.1 *m* is the message to be transferred, *c* is the encoded message sent via the channel, *c'* is the received message, and *m'* is an estimate of the original message.
The two most important goals of information transmission:

- get the message to the sink with as little distortion as possible (where the error is introduced by the channel's noise)

- use the channel as efficiently as possible (as a price has to be paid for the usage)

To achieve high quality information transmission, we introduce a **decoder** and an **encoder**. The former at the beginning of the channel, the latter at the end. Both are designed according to the previous two goals.

A message is interpreted as a series of symbols emitted by a source. We can distinguish between two types of symbol sets: the **discrete** (e.g. letters) and the **continuous** (e.g. vocal chords) ones. Based on how the symbols are distributed in time we can also identify two groups: **discrete** (e.g. written text) and **continuous** (e.g. speech) cases.
For both continuous categories we can produce the discrete equivalents by applying the proper sampling and quantizing methods (e.g. writing down a conversation to a piece of paper). Because of this property the field of information theory deals primarily with sources of discrete-time and discrete symbols sets, and so does this syllabus from now on.

A source can be characterised by the symbols set it generates (**source alphabet**) and the corresponding probabilities of how frequently each symbol is emitted (**source distribution**):

$$A = \{a_0, a_1, \dots, a_{M-1}\},$$
$$P(A) = \{p_0, p_1, \dots, p_{M-1}\}, \sum_i p_i = 1\,, p_i \neq 0$$

where *M* is the cardinality of the symbol set A (the number of its elements, also written as |A|), and the sum of the probabilities is 1 (i.e. the source always emits a member of its alphabet). Apart this, the sources mentioned in this syllabus are (in most cases) expected to also be

- **stationary** – the probabilities are independent of time
- **memoryless** – the symbols (and thus their probabilities) are statistically independent from each other

The "amount of surprise", i.e. the information content of the occurrence of the i-th symbol can be defined as

$$I(a_i) = \log\frac{1}{p_i}\ [\text{bit}]$$

The unit of information is **bit** (binary unit)[3], because a logarithm of base 2 is used. It can be seen that the information contents of statistically independent events are to be summed.
The **entropy** of a source is *the average amount of information conveyed per source symbol*:

$$H(A) = \sum_i p_i \log\frac{1}{p_i}\ [\text{bit}]$$

For example, a source P(A) = {0.5, 0.5} has an entropy of 1 bit. If we allow the values of symbol probability to be 0 or 1, a source P(A) = {0, 1} can be constructed that has the entropy of 0 bit[4] (but it is quite a boring source as it emits the same symbol over and over again). It can be shown that equal distribution maximises entropy and this maximum value is log(*M)* (see Appendix A1). From this we can conclude that for any source with M number of symbols $0 \leq H(A) \leq \log(M)$.

An example: for source P(A) = {0.4, 0.6}, H(A) = 0.970 bits. From now on, instead of P(A) = {…}, a shorter notation A = {…} will be used if the specific symbols are not that important in the problem.

[3] Not to be mistaken as the digits of binary numbers, which are called the same. The connection between these two is that the information content of a digit of a binary number can be *at most* 1 bit.

[4] because $\lim_{x\to 0} x \log\frac{1}{x} = 0$

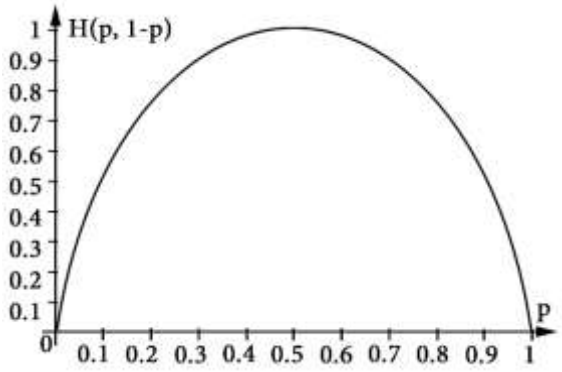


Figure 1.2: *The connection between a source's entropy and distribution*

1.3. Conditional Entropy and Mutual Information

These two concepts are used to describe the mutual, average information content of two or more sources. They will also appear in the coding methods related section.

Suppose we have two sources A and B, e.g. the current images displayed on two separate TV screens. Let $p_{i,j}$ denote the probability of source A emitting the i-th and B the j-th symbol. Then the **mutual (joint) entropy** and the **conditional entropy** can be defined as

$$H(A,B) = \sum_{i,j} p_{i,j} \log \frac{1}{p_{i,j}}$$

$$H(B|A) = \sum_{i} p_i \sum_{j} p_{j|i} \log \frac{1}{p_{j|i}}$$

where $p_{j|i}$ might be 0, for this case see footnote #4. The joint entropy H(A, B) is the average amount of information conveyed by both sources together. It can be seen that if the sources are independent, then H(A, B) is the sum of the individual entropies.
If we expand the definition of H(A ,B) and we apply $p_{i,j} = p_i p_{j|i}$, we get the equation

$$H(A,B) = H(A) + H(B|A)$$

where H(B|A) is the entropy of B given A, the conditional entropy[5]. This shows how much *additional* information we gain from B on average, if we already know source A. From the equation it can be seen that H(B|A) conditional entropy is nothing else but the weighted average of the entropies computed form the conditional source distributions that belong to events (symbols) in A. Of course, it is also true that H(A, B) = H(B) + H(A|B).

If source B is **completely independent** of A (e.g. one TV shows football and the other displays romantic series) than we expect H(A, B) joint entropy to be equal to the sum of the individual entropies of the sources (our knowledge is increased from both sources individually). Indeed, based on the statistical independence of the symbol probabilities the conditional probabilities become "unconditional", thus H(B|A) turns into H(B) and we get H(A, B) = H(A) + H(B).

On the other hand, if B is **completely dependent** on A then we expect no additional information from B (if both TVs show the same then we would know exactly the same if we had only one device). Which means H(A, B) = H(A). Evidently, because B is completely dependent, all conditional probabilities $p_{j/i}$ are either 0 or 1, so H(B|A) = 0.
Therefore, we can state that

[5] It is easier to memorise this equation if we notice its similarity to the connection between conditional and joint probabilities, except that because of the logarithmic definition of information, here we have addition instead of multiplication.

$$0 \leq H(B|A) \leq H(B),$$

i.e. *side information does not increase entropy*. An example is the parity bit, a completely defined source that has less entropy (Example 1.1).

The **mutual information I(B, A)** of sources A and B can be defined as that part of the average information content of B which is provided by source A. To get this we need to subtract from H(B) the part that is *not* conveyed in A, i.e. the average "surprise" caused by B compared to A, which is nothing else but H(B|A):

$$I(B,A) = H(B) - H(B|A)$$

It can be easily seen that because of the two possible interpretation of H(A, B)

$$I(B,A) = I(A,B)$$

If the two sources are **completely independent**[6] of each other, then conditional entropy H(B|A) turns into H(B), thus the mutual information is 0, what it is expected to be. However, if B is completely dependent on A then H(B|A) = 0, so I(B, A) = H(B), i.e. the complete H(B) is contained in source A. But this does not imply that H(A) = H(B). For example, if A has more symbols than B (|A| > |B|), then H(A) > H(B), so B has less average information content. So in the case when such an A is a result of an observation at the beginning of a communication process and B is generated at the end, then we can conclude that some information is lost during the transmission.

The same is true for every situation in which H(B) is not smaller than H(A) yet I(B, A) < H(A) because some part of the information in A "did not pass through". A classic example for this is the *town gossip*. The story we are told is usually more interesting (it's more surprising, than the source) than the original/true version, but important parts of the truth (the original information content) are usually lost in the transmission… So the question arises: where the additional information is coming from? The answer: from the channel.

[6] The independence of two sources can be deduced from the problem's description: e.g. the evening prayer of the imam in Istanbul (source A) and a traffic light with faulty controlling sequences in Veszprém (source B) can be considered to be independent.

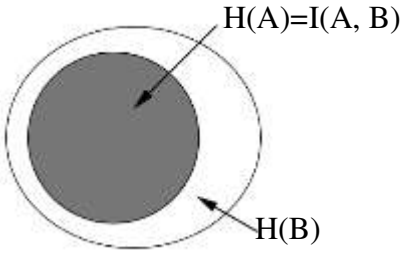
Example 1.1

Some basic examples for information transmission. We suppose that the information goes from A to B. The mutual information is greyed in. Examples 1. and 3. have discrete, 2., 4. and 5. have continuous values.

1. The case of the telegraphist with Parkinson's disease, whose hand is shaking (no information is lost, but additional symbols are added that are typical to the telegraphist, not to the message):

Source A: the message given by the captain, in Morse code

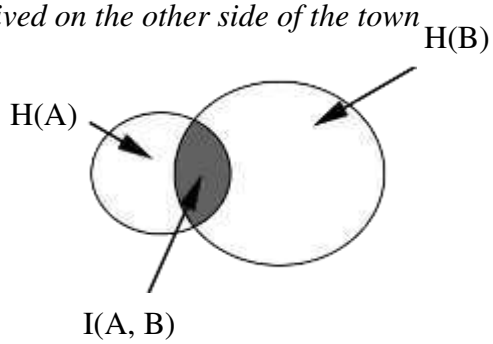
Source B: the message received on the other ship



2. Town gossip (information is lost, and new components are added):

Source A: the original news

Source B: the news received on the other side of the town

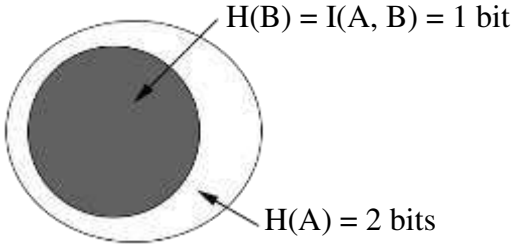


H(A) < H(B)

3. 2+1 parity bit (completely defined source, some information is lost):

Source A: A = {a0, a1, a2, a3} = {0.25, 0.25, 0.25, 0.25}

Source B: B = {b0, b1}. B *emits* b0 *if* A *produces* a0 *or* a2*, otherwise it sends* b1.

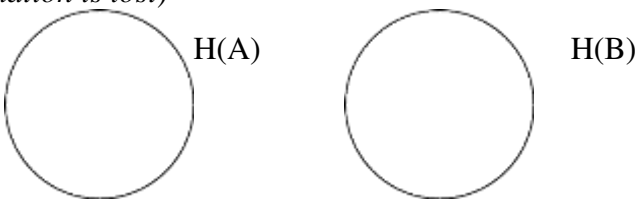


4. Perfect translation (completely defined source, no information loss):
Source A: this syllabus in Hungarian
Source B: this syllabus in English



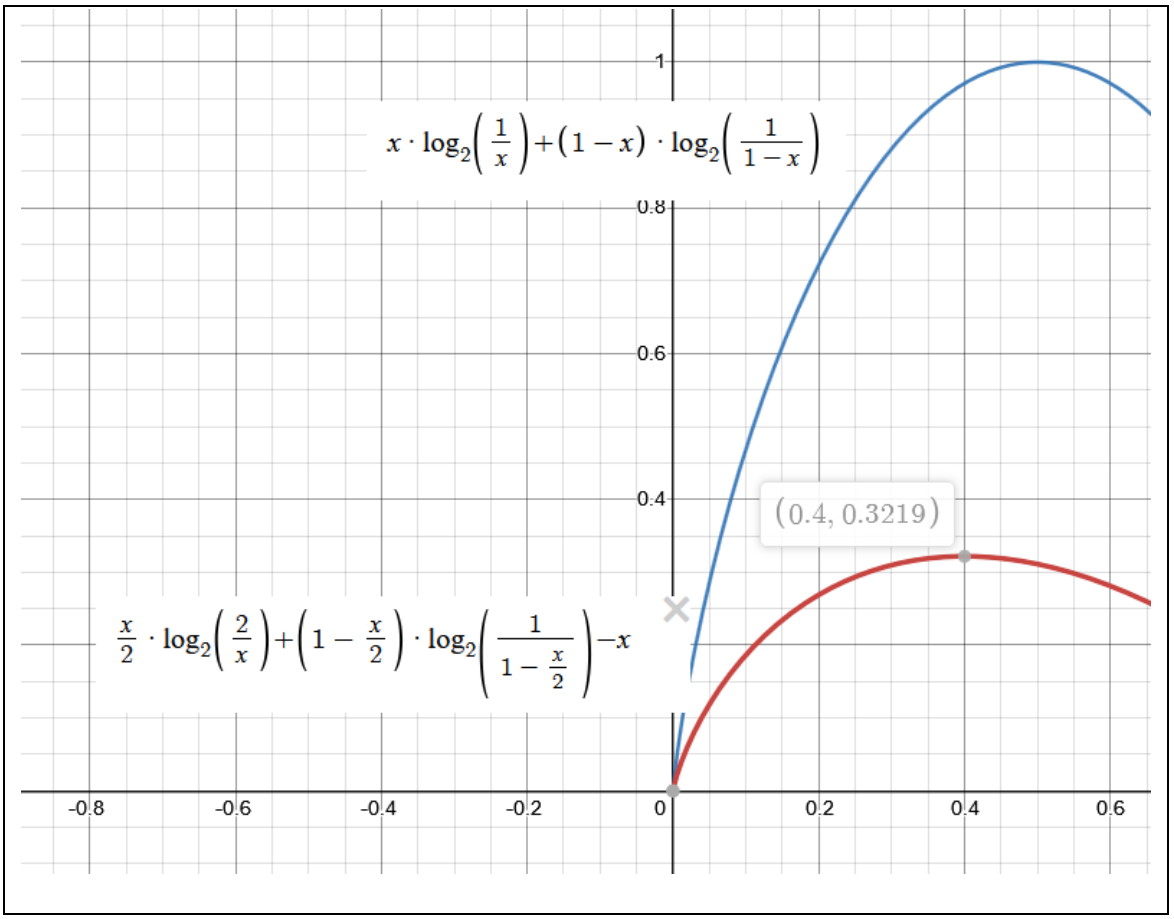
H(A) = I(A, B) = H(B)

5. Independent sources:
Source A: the prayer of an imam in Istanbul
Source B: a traffic light with faulty controlling sequences in Veszprém
(*independent sources – if they describe the same process, but at different time moments, then it means all information is lost)*



I(A, B) = 0

6. The information content of the fast antigen (AG) Covid test:
We know that:
--The AG test can detect the Covid virus in 50% of the cases.
--If the antigen test is positive, then the patient is for sure Covid positive (100%)
So if the probability of the patient being Covid-positive is p, then we have C = {p, 1-p} *as the source that we want to predict based on the AG result.*
How much information can we gain from the AG about C, in the function of p?
AG = {p/2, 1-p/2}
I(C, AG) = H(AG) - H(AG|C). *The maximum is about 0.32 bit, when p=0.4 (see below).*
In other words, about ½..1/3 of the information content of C is accessible by observing AG, depending on p, for p < ½.


$x \cdot \log_2\left(\frac{1}{x}\right)+(1-x) \cdot \log_2\left(\frac{1}{1-x}\right)$
$\frac{x}{2} \cdot \log_2\left(\frac{2}{x}\right)+\left(1-\frac{x}{2}\right) \cdot \log_2\left(\frac{1}{1-\frac{x}{2}}\right)-x$
(0.4, 0.3219)
1
0.8
0.6
0.4
-0.8
-0.6
-0.4
-0.2
0
0.2
0.4
0.6